ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 2

ELECTRONICS

PAPER 1 · PRACTICE A · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central NC(V) Level 2 Electronics paper 1 · practice a.

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How to mark: each ✓ is one mark. Give marks for the formula, the substitution and the answer with the correct unit. Use consistent accuracy: if a wrong answer is carried forward correctly, give the marks for the later steps. Accept other correct answers.

SECTION A

QUESTION 1

1.1.1 D ✓
1.1.2 B ✓
1.1.3 D ✓
1.1.4 B ✓
1.1.5 C ✓
1.1.6 A ✓
1.1.7 B ✓
1.1.8 D ✓
1.1.9 A ✓
1.1.10 C ✓

(10)

1.2.1 True ✓
1.2.2 False ✓
1.2.3 True ✓
1.2.4 False ✓
1.2.5 True ✓
1.2.6 False ✓
1.2.7 True ✓
1.2.8 True ✓
1.2.9 True ✓
1.2.10 False ✓

(10)

1.3.1 D ✓
1.3.2 J ✓
1.3.3 F ✓
1.3.4 E ✓
1.3.5 A ✓
1.3.6 H ✓
1.3.7 B ✓
1.3.8 I ✓
1.3.9 C ✓
1.3.10 G ✓

(10)

TOTAL SECTION A: [30]

SECTION B

QUESTION 2

2.1 Any FOUR: switch off and unplug equipment before working on it ✓; wear an anti-static wrist strap ✓; do not work with wet hands or near water ✓; use insulated tools ✓; never open a power supply unit or monitor; report damaged cables; no food or drink in the workshop; keep the work area tidy (4)

2.2 A conductor allows electric current to flow easily (has many free electrons) ✓, e.g. copper ✓; an insulator does not allow current to flow easily (few free electrons) ✓, e.g. plastic / rubber / glass ✓. (4)

2.3 DC flows in one direction only with a constant value ✓ (e.g. a battery); AC changes direction (polarity) regularly ✓ (e.g. the mains supply). Waveforms: DC – a straight horizontal line above the zero axis ✓✓; AC – a sine wave above and below the zero axis ✓✓. (6)

2.4 Any TWO with examples: heating effect ✓ – kettle, iron, soldering iron ✓; magnetic effect ✓ – electric motor, relay, speaker ✓; chemical effect – charging a battery, electroplating; lighting effect – lamp, LED (4)

2.5 Switch off the supply (at the main switch / unplug) ✓ or move the person away using a dry, non-conducting object such as a wooden broom; do not touch the person with bare hands ✓. (2)

[20]

QUESTION 3

3.1 The current through a conductor is directly proportional to the voltage across it ✓, provided the temperature (resistance) stays constant ✓. (2)

3.2 $I = V / R$ ✓ $= 5 / 2{,}5$ ✓ $= 2$ A ✓ (3)

3.3 (a) $R = V / I$ ✓ $= 230 / 0{,}25$ ✓ $= 920\ \Omega$ ✓ (b) $P = V \times I$ ✓ $= 230 \times 0{,}25$ ✓ $= 57{,}5$ W ✓ (6)

3.4 $E = P \times t = 0{,}06$ kW $\times (5 \times 30)$ h ✓ $= 0{,}06 \times 150 = 9$ kWh ✓; cost $= 9 \times$ R3,20 ✓ $=$ R28,80 ✓ (4)

[15]

QUESTION 4

4.1.1 $1/R_p = 1/6 + 1/3$ ✓ $= 3/6$ ✓; $R_p = 2\ \Omega$ ✓ (or $R_p = (6 \times 3)/(6 + 3) = 2\ \Omega$) (3)

4.1.2 $R_T = R_1 + R_p = 4 + 2$ ✓ $= 6\ \Omega$ ✓ (2)

4.1.3 $I_T = V / R_T = 12 / 6$ ✓ $= 2$ A ✓ (2)

4.1.4 $V_1 = I \times R_1 = 2 \times 4 \checkmark = 8\ V \checkmark$ (2)

4.1.5 $V_3 = V - V_1 = 12 - 8 \checkmark = 4\ V \checkmark (\text{or } 2 \times 2\ \Omega)$ (2)

4.1.6 $I_2 = V_2 / R_2 = 4 / 6 \checkmark = 0{,}67\ A \checkmark$ (2)

4.1.7 $P = V \times I = 12 \times 2 \checkmark = 24\ W \checkmark$ (2)

4.2 Any TWO: each light receives the full supply voltage (full brightness) ✓; if one lamp fails, the others keep working ✓; each light can be switched on and off independently (2)

4.3 (a) 3 × 1,5 = 4,5 V ✓ (b) 1,5 V ✓ (2)

[19]

QUESTION 5

5.1 Yellow = 4, violet = 7, orange = × 1 000 ✓ → 47 000 Ω = 47 kΩ ✓; gold = ± 5% tolerance ✓ (3)

5.2 Correct standard symbols: (a) diode – triangle pointing to a bar ✓ (b) capacitor – two parallel plates ✓ (c) fuse – rectangle with a line through it (or S-curve) ✓ (d) LED – diode symbol with two arrows pointing outwards ✓ (4)

5.3 The resistor limits the current through the LED ✓ so that the LED does not burn out / is not damaged ✓. (2)

5.4 Switch off and remove the fuse ✓; set the multimeter to the resistance (Ω) or continuity range ✓; place the probes on both ends – a reading of about 0 Ω (or a beep) means it is good, an open circuit (OL / infinite) means it is blown ✓. (3)

5.5 Any TWO: smoothing (filtering) the output of a power supply ✓; storing charge / energy (e.g. a camera flash) ✓; timing circuits; blocking DC while passing AC (coupling); keeping memory powered for a short time (2)

5.6 (a) CPU / RAM / chipset (millions of transistors) ✓ (b) motherboard / power supply unit / graphics card ✓ (2)

[16]

TOTAL SECTION B: [70]

TOTAL: 100

END OF MARKING GUIDELINE